

CHURCHILL, MB, Canada

WMO: 716180

Lat: 58.740N

Lon: 94.070W

Elev: 29

StdP: 100.97

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.0	-35.3	-41.7	0.1	-36.8	-40.0	0.1	-35.3	15.8	-22.6	14.4	-23.2	5.3	290	0.960

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	25.7	17.4	23.2	16.0	20.8	14.9	18.3	23.8	16.9	21.7	15.4	19.9	5.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.2	11.5	20.8	14.6	10.4	19.3	13.2	9.5	17.8	52.0	23.9	47.6	21.7	43.4	19.8	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.5	12.9	11.4	-39.0	30.1	2.2	1.4	-40.6	31.1	-41.9	32.0	-43.1	32.8	-44.7	33.8		
			-39.1	21.3	2.2	1.0	-40.7	22.0	-41.9	22.6	-43.2	23.1	-44.7	23.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-5.7	-25.3	-24.0	-18.5	-9.5	-0.9	7.8	13.2	12.5	7.3	-0.2	-11.6	-20.5	(6)
(7)		DBStd	14.87	6.58	6.53	7.56	7.10	5.54	5.28	4.37	3.50	3.95	4.03	6.71	7.36	(7)
(8)		HDD10.0	5996	1093	953	883	586	341	104	18	11	99	317	647	944	(8)
(9)		HDD18.3	8787	1351	1186	1141	836	596	317	169	184	330	575	897	1203	(9)
(10)		CDD10.0	269	0	0	0	0	3	39	116	90	20	1	0	0	(10)
(11)		CDD18.3	17	0	0	0	0	0	2	9	4	1	0	0	0	(11)
(12)		CDH23.3	197	0	0	0	0	2	43	107	43	3	0	0	0	(12)
(13)		CDH26.7	38	0	0	0	0	1	8	22	8	1	0	0	0	(13)
(14)	Wind	WSAvg	5.8	6.0	6.1	5.9	5.9	5.7	5.2	4.9	5.3	6.0	6.3	6.2	5.8	(14)
(15)	Precipitation	PrecAvg	449	20	16	20	26	38	49	57	67	61	52	39	22	(15)
(16)		PrecMax	689	62	38	43	78	130	140	132	130	160	105	98	47	(16)
(17)		PrecMin	311	4	2	3	1	4	11	11	15	20	12	11	1	(17)
(18)		PrecStd	97	12	10	11	21	28	33	24	25	29	22	19	11	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.4	-3.5	3.7	9.7	21.6	27.6	29.2	27.9	22.7	13.1	2.6	-2.5	(19)
(20)			MCWB	-5.9	-4.6	1.4	5.2	13.1	16.6	18.9	19.0	15.7	9.0	1.5	-3.0	(20)
(21)		2%	DB	-9.1	-8.4	0.2	5.5	14.6	24.0	26.4	24.0	18.8	8.5	0.4	-4.6	(21)
(22)			MCWB	-9.4	-9.1	-1.5	2.3	9.3	15.2	18.0	17.1	13.9	6.3	-0.1	-5.3	(22)
(23)		5%	DB	-12.4	-11.3	-3.4	2.9	10.2	20.9	23.7	21.7	15.6	6.2	-0.8	-6.6	(23)
(24)			MCWB	-12.8	-12.0	-4.5	0.8	6.6	13.5	16.4	16.0	12.1	4.9	-1.4	-7.2	(24)
(25)		10%	DB	-15.4	-13.8	-7.0	0.6	6.5	17.4	21.1	18.9	13.1	4.5	-2.2	-9.0	(25)
(26)			MCWB	-15.8	-14.4	-7.8	-0.8	4.1	12.3	15.4	14.7	10.7	3.6	-2.8	-9.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-5.8	-4.5	1.6	5.6	13.6	18.4	20.6	20.2	16.7	9.3	1.6	-2.9	(27)
(28)			MCDB	-5.5	-3.7	3.7	9.2	20.6	25.1	27.1	26.1	21.1	13.0	2.5	-2.5	(28)
(29)		2%	WB	-9.4	-9.1	-1.4	2.5	9.5	15.6	18.6	18.1	14.4	6.6	0.1	-5.3	(29)
(30)			MCDB	-9.0	-8.4	0.1	5.2	14.4	22.5	24.4	23.1	18.2	8.1	0.4	-4.7	(30)
(31)		5%	WB	-12.8	-11.9	-4.5	0.9	6.7	14.0	17.2	16.5	12.4	5.1	-1.4	-7.1	(31)
(32)			MCDB	-12.5	-11.4	-3.4	2.7	10.1	20.5	22.3	20.6	14.9	6.0	-1.0	-6.6	(32)
(33)		10%	WB	-15.8	-14.4	-7.8	-0.7	4.1	12.3	15.9	15.0	10.7	3.7	-2.8	-9.4	(33)
(34)			MCDB	-15.4	-13.9	-7.0	0.6	6.5	17.5	20.4	18.4	12.8	4.4	-2.2	-9.0	(34)
(35)	Mean Daily Temperature Range	MDBR	7.3	8.0	9.2	9.0	7.0	9.8	9.6	7.8	6.3	4.6	6.8	7.2	(35)	
(36)		5% DB	MCDBR	9.1	10.7	12.2	12.2	14.2	16.5	14.7	13.3	11.6	7.5	5.8	8.6	(36)
(37)			MCWBR	9.0	10.3	10.9	9.6	9.7	9.2	7.6	7.3	7.3	5.7	5.6	8.4	(37)
(38)		5% WB	MCDBR	9.0	10.6	12.0	11.5	13.8	15.8	13.3	12.2	10.7	6.9	5.6	8.4	(38)
(39)			MCWBR	9.0	10.2	10.9	9.2	9.5	9.6	7.8	7.3	7.3	5.5	5.5	8.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.208	0.251	0.278	0.317	0.336	0.339	0.361	0.339	0.310	0.290	0.232	0.185	(40)	
(41)		taud	2.126	2.159	2.155	2.091	2.173	2.344	2.442	2.524	2.578	2.486	2.211	2.046	(41)	
(42)		Ebn at Noon	652	784	865	884	891	895	866	865	843	749	606	548	(42)	
(43)		Edh at Noon	46	76	103	132	131	113	101	87	70	56	42	34	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.59	1.60	3.22	4.99	5.35	5.99	5.40	4.06	2.50	1.14	0.51	0.34	(44)	
(45)		RadStd	0.05	0.11	0.26	0.46	0.58	0.54	0.35	0.32	0.24	0.12	0.07	0.03	(45)	

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page